Subnet Mask Documentation

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1. Introduction

A subnet mask is a number used in networking to divide an IP address into two parts: the network part and the host (device) part. It helps devices understand how to communicate inside a network.
In simple words, a subnet mask tells which part of an IP address represents the network and which part represents the device.

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2. What is an IP Address

An IP address is a unique number given to every device connected to a network.

Example:

192.168.1.1
This IP address is divided into four parts:
192 | 168 | 1 | 10

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3. What is Subnet Mask

A subnet mask looks like an IP address and is written in the same format.

Example:

255.255.255.0
It works together with the IP address to identify:

- Network (area)
- Host (device)

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4. How Subnet Mask Works

Example:

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Now match both:

IP: 192 . 168 . 1 . 10
Mask: 255 . 255 . 255 . 0

Rule:

255 = Network part
0 = Host part

So:

192.168.1 = Network
10 = Device (Host)

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5. Easy Trick to Remember

- Where subnet mask is 255 → Network
- Where subnet mask is 0 → Host

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6. Common Subnet Masks

255.0.0.0 → Large network
255.255.0.0 → Medium network
255.255.255.0 → Small network (most common)
255.255.255.128 → Very small network

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7. Why Subnet Mask is Used

Subnet mask is used for:

Identifying network and host
Checking if two devices are in the same network
Controlling communication
Managing network efficiently

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8. Real Life Example

Think of an address:

Village + House Number

Village = Network
House Number = Host

Example:

192.168.1 → Village
10 → House

Subnet mask tells where village ends and house number starts.

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9. Communication Example

Device 1: 192.168.1.10
Device 2: 192.168.1.20
Subnet Mask: 255.255.255.0

Both are in same network → Direct communication

Device 3: 192.168.2.10
Different network → Router is required

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10. Where Subnet Mask is Used

- Routers (to route traffic)
- Computers, laptops, mobiles
- Cloud networks (AWS VPC, Subnets)

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11. How Subnet Mask is Assigned

Subnet mask can be:

- Manually set by network admin
- Automatically given by router (DHCP)

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12. Important Concept

IP Address = Identity of device
Subnet Mask = Rule to divide network and host

Both work together for communication.

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13. Final Definition

A subnet mask is a number that divides an IP address into network and host parts, helping devices communicate correctly within a network.

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14. One-Line Revision

Subnet mask is used to identify network and device parts of an IP address.